

BESS Financial Analysis

Anarita 10 MW

Battery Energy Storage System — Investment Return Analysis

CURTAILMENT RECOVERY & REVENUE MODELLING

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Prepared for Muminjon / Spanercom

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Executive Summary

This analysis evaluates three BESS configurations for the Anarita 10 MW PV park (Fixed Tilt, 19,324 MWh/yr PVSyst verified), modelling curtailment recovery revenue across five curtailment scenarios using verified TSOC market data and 18 months of actual park curtailment records (May 2024 – Oct 2025).

34.7%

BEST YEAR-1 ROI
(OPTION A)

2.9 yr

FASTEST PAYBACK
(OPTION A)

€183/MWh

PEAK EVENING PRICE
(17:00–21:00)

48%

2025 CURTAILMENT
RATE (ACTUAL)

Key Finding

CURRENT CURTAILMENT (48%
ACTUAL)

Option A

20 MWh — 34.7% ROI
Highest return, fastest payback

EXPECTED RANGE (55–65%)

Option B

30 MWh — 33.2–38.2% ROI
Best balance of return & capacity

HIGH CURTAILMENT (70%+)

Option C

40 MWh — 33.9% ROI
Maximum energy capture

Recommendation

Anarita's actual measured curtailment has risen from 18% (summer 2024) to 48% (2025 average), with spring peaks of 66–67%. At this level, **Option A (20 MWh)** delivers the highest ROI at 34.7% with a 2.9-year payback. However, with curtailment accelerating rapidly — and the park already hitting 66–67% in spring 2025 — **Option B (30 MWh)** provides better long-term value. If curtailment averages 55%+ (which current park data strongly suggests is imminent), Option B captures significantly more revenue while maintaining ROI above 33%. Option C (40 MWh) becomes the optimal choice only if curtailment exceeds 70%.

Park Profile — Anarita 10 MW PV

Location	Anarita, Paphos District	Client	Muminjon / Spanercom
Installed Capacity	10 MW _p	EPC Provider	Lighthief Cyprus Ltd
Configuration	Fixed Tilt (EAST + WEST arrays)	BESS OEM	Linyang Energy
Annual Production (PVSyst)	19,324 MWh/yr	Market	Day-Ahead (TSOC)
Specific Yield	1,932 kWh/kWp	Revenue Model	Curtailment Recovery
Data Period	May 2024 – Oct 2025 (18 months verified)	Analysis Date	March 2026
2025 Avg Curtailment	48%		
PV Degradation	0.5%/yr		
Grid Connection	TSO Cyprus (EAC)		

Why BESS for a 10 MW PV Park?

Cyprus's electricity grid cannot absorb all solar generation during midday hours, resulting in mandatory curtailment — energy the park produces but cannot sell. A co-located BESS captures this otherwise-lost energy, stores it, and discharges during peak evening hours when prices are highest. The cost of charging is effectively €0/MWh since the energy would otherwise be curtailed (wasted).

Cyprus Curtailment Context

The Curtailment Trajectory

Solar curtailment on the Cyprus grid has risen exponentially as PV capacity outpaces grid infrastructure and demand:

Year	Curtailment Rate	Trend	Context
2021	~0%		Minimal solar penetration
2022	~8%		First curtailment events
2023	~22%		Rapid solar build-out
2024	~35%		Grid saturation increasing
2025	48%		Actual (Anarita park, Jan–Oct 2025)
2026–27	55–65%		Expected — more PV online, EuroAsia delay

Key Insight

Curtailment is not cyclical — it is structurally increasing. Every new MW of solar added to Cyprus's isolated grid pushes curtailment higher. The EuroAsia Interconnector (to Greece/Israel) is delayed to 2029+. Without grid-scale storage, curtailment above 60% is likely within 12–18 months.

Monthly Electricity Price Patterns (Verified TSOC Data)

134-day dataset from October 2025 to February 2026:

Month	Peak (17:00–21:00)	Midday (10:00–14:00)	Spread	Solar Mix %
Oct 2025	€177.69	€78.89	€98.80	27.9%
Nov 2025	€184.77	€81.50	€103.27	22.5%
Dec 2025	€147.55	€98.30	€49.25	17.1%
Jan 2026	€188.26	€121.26	€67.00	15.2%
Feb 2026	€191.88	€84.75	€107.13	22.3%
Average	€182.99	€101.13	€81.86	21.0%

Source: TSOC (Transmission System Operator Cyprus) Day-Ahead Market data, 134 consecutive days.

Anarita Park — Actual Measured Curtailment

18 months of verified meter data from the Spanercom Anarita EAST + WEST arrays (PVSyst simulation cross-referenced with DSO dispatch-down records):

Month	Energy Yield (MWh)	Potential (MWh)	Curtailed %
May-24	1,023	1,854	45%
Jun-24	1,670	1,888	12%
Jul-24	1,791	1,916	7%

Aug-24	1,703	1,788	5%
Sep-24	1,225	1,589	23%
Oct-24	941	1,617	42%
Nov-24	645	1,160	44%
Dec-24	694	951	27%
Jan-25	600	1,179	49%
Feb-25	655	1,198	45%
Mar-25	566	1,648	66%
Apr-25	578	1,731	67%
May-25	850	1,769	52%
Jun-25	1,376	1,898	28%
Jul-25	1,438	1,968	27%
Aug-25	1,147	1,878	39%
Sep-25	965	1,725	44%
Oct-25	467	1,139	59%
2025 Average (Jan–Oct)			48%

Trend Alert

Curtailment has risen from 18% (summer 2024 average) to 48% (2025 average), with peaks of 66–67% in spring. This is an accelerating trend driven by continued solar build-out on Cyprus's isolated grid with no new demand or interconnection capacity. Source: PVSyst verified (May 2024 – Oct 2025), DSO dispatch-down records for Spanercom Anarita EAST + WEST arrays.

How Curtailment Recovery Works

Under current Cyprus regulations (as of February 2026), BESS systems co-located with PV parks cannot purchase energy from the Day-Ahead Market grid. Revenue is derived exclusively from recovering energy that would otherwise be curtailed.



Revenue Per MWh Stored



The Economics Are Asymmetric

Because the charge cost is €0 (the energy would otherwise be wasted), every MWh stored and discharged at peak generates €157.96 in net revenue (€182.99 × 86.32% RTE). This is fundamentally different from grid arbitrage, where you must pay the midday price to charge.

BESS Technical Parameters

OEM / System	Linyang Energy — Full Turnkey BESS
Chemistry	LFP (Lithium Iron Phosphate)
Round-Trip Efficiency	86.32% (AC-to-AC, full system)
Cycling Regime	1 cycle per day
C-Rate	0.25P (4-hour duration at rated capacity)
Availability (LTSA)	97% guaranteed
SOH Year 1	94.62%
SOH Year 10	79.58%
SOH Year 15	73.61%

BESS Options Comparison

Option A

10 MW / 20 MWh

€3,033,000

€151.65 / kWh

Duration

2 hours

E:P Ratio

2:1

Annual O&M

€64,392

Best at

45–55% curtailment

Option B

10 MW / 30 MWh

RECOMMENDED

€4,040,500

€134.68 / kWh

Duration

3 hours

E:P Ratio

3:1

Annual O&M

€89,088

Best at

55–65% curtailment

Option C

10 MW / 40 MWh

€5,048,000

€126.20 / kWh

Duration

4 hours

E:P Ratio

4:1

Annual O&M

€113,784

Best at

70%+ curtailment

O&M Cost Breakdown (LTSA Tier C, OEM-Confirmed)

Component	Rate (€/MWh/yr)	20 MWh	30 MWh	40 MWh
BESS (cells, racks, thermal)	€1,157.62	€23,152	€34,729	€46,305
PCS + MV Equipment	€1,311.97	€26,240	€39,359	€52,479
Subtotal LTSA	€2,469.59	€49,392	€74,088	€98,784
Insurance (annual)	—	€15,000	€15,000	€15,000
Total Annual O&M	—	€64,392	€89,088	€113,784

Curtailment Scenario Matrix — Year 1

All three BESS options modelled across five curtailment scenarios. The annual PV production is 19,324 MWh (PVSyst verified). Peak discharge price: €182.99/MWh. Charge cost: €0/MWh (curtailment recovery).

Curtailment	45%				50%				55%			
	Revenue	Net	ROI	Payback	Revenue	Net	ROI	Payback	Revenue	Net	ROI	Payback
A — 20 MWh	€1.12M	€1.05M	34.7%	2.9 yr	€1.12M	€1.05M	34.7%	2.9 yr	€1.12M	€1.05M	34.7%	2.9 yr
B — 30 MWh	€1.09M	€1.00M	24.8%	4.0 yr	€1.30M	€1.21M	30.0%	3.3 yr	€1.43M	€1.34M	33.2%	3.0 yr
C — 40 MWh	€1.09M	€0.98M	19.4%	5.2 yr	€1.21M	€1.10M	21.8%	4.6 yr	€1.34M	€1.22M	24.2%	4.1 yr

Curtailment	60%				70%			
	Revenue	Net	ROI	Payback	Revenue	Net	ROI	Payback
A — 20 MWh	€1.12M	€1.05M	34.7%	2.9 yr	€1.12M	€1.05M	34.7%	2.9 yr
B — 30 MWh	€1.63M	€1.55M	38.2%	2.6 yr	€1.68M	€1.59M	39.3%	2.5 yr
C — 40 MWh	€1.46M	€1.34M	26.6%	3.8 yr	€1.82M	€1.71M	33.9%	3.0 yr

Green cells indicate the best-performing option for each curtailment scenario.

* Option A revenue is constant across all curtailment scenarios because the 20 MWh battery completes one full cycle per available day (354 of 365). Revenue is independent of curtailment rate — a strength for predictability but a limitation for upside capture.

Detailed Year 1 Numbers

Curtailment	Option	Captured (MWh)	Discharged (MWh)	Revenue	O&M	Net Income	ROI	Payback
45% 8,696 MWh	A — 20 MWh	7,081	6,112	€1,118,493	€64,392	€1,054,101	34.7%	2.9 yr
	B — 30 MWh	6,915	5,971	€1,092,573	€89,088	€1,003,485	24.8%	4.0 yr
	C — 40 MWh	6,915	5,971	€1,092,573	€113,784	€978,789	19.4%	5.2 yr
50% 9,662 MWh	A — 20 MWh	7,081	6,112	€1,118,493	€64,392	€1,054,101	34.7%	2.9 yr
	B — 30 MWh	8,247	7,119	€1,302,691	€89,088	€1,213,603	30.0%	3.3 yr
	C — 40 MWh	7,683	6,632	€1,213,871	€113,784	€1,100,087	21.8%	4.6 yr
55% 10,628 MWh	A — 20 MWh	7,081	6,112	€1,118,493	€64,392	€1,054,101	34.7%	2.9 yr
	B — 30 MWh	9,071	7,830	€1,432,846	€89,088	€1,343,758	33.2%	3.0 yr
	C — 40 MWh	8,451	7,295	€1,335,001	€113,784	€1,221,217	24.2%	4.1 yr
	A — 20	7,081	6,112	€1,118,493	€64,392	€1,054,101	34.7%	2.9 yr

60% 11,594 MWh	MWh							
	B — 30 MWh	10,346	8,931	€1,634,203	€89,088	€1,545,115	38.2%	2.6 yr
	C — 40 MWh	9,220	7,959	€1,456,467	€113,784	€1,342,683	26.6%	3.8 yr
70% 13,527 MWh	A — 20 MWh	7,081	6,112	€1,118,493	€64,392	€1,054,101	34.7%	2.9 yr
	B — 30 MWh	10,622	9,169	€1,677,735	€89,088	€1,588,647	39.3%	2.5 yr
	C — 40 MWh	11,543	9,965	€1,823,488	€113,784	€1,709,704	33.9%	3.0 yr

Optimal Size Recommendation

Decision Framework

Scenario	Curtailment Range	Best Option	ROI Range	Payback	Rationale
Conservative	45–50%	Option A (20 MWh)	34.7%	2.9 yr	Lowest capex, highest ROI. BESS fully utilized every day.
Expected	55–65%	Option B (30 MWh)	33.2–38.2%	2.6–3.0 yr	Best balance. Captures more revenue as curtailment rises.
Aggressive	70%+	Option C (40 MWh)	33.9%+	3.0 yr	Maximum capture. Highest absolute revenue at extreme curtailment.

Our Recommendation: Option B — 10 MW / 30 MWh (€4,040,500)

Option A delivers the best ROI at 34.7%, but its 20 MWh capacity is already fully utilized at the park's current 48% curtailment — meaning it cannot capture additional revenue as curtailment increases. Option B starts with a strong 33.2% ROI at 55% curtailment and improves to 38.2% at 60%, reaching 39.3% at 70%, capturing up to €535K more annual revenue than Option A. With Anarita's curtailment already hitting 66–67% in spring 2025 and averaging 48% for the year, the additional €1M investment in Option B is recovered in under 2 years from the incremental revenue alone.

Why Not Option A?

Ceiling Risk

Option A's 20 MWh battery is "full" every day at the park's current 48% curtailment. With curtailment already reaching 66–67% in spring 2025, Option A leaves up to €535K/yr of recoverable revenue on the table — energy that is curtailed but the battery is too small to capture. Over 15 years, this represents €6–8M in lost revenue.

Why Not Option C?

Underutilization at Current Levels

Option C's 40 MWh battery requires 70%+ curtailment to approach full utilization. At 45% curtailment, it delivers 19.4% ROI with a 5.2-year payback — the additional 10 MWh over Option B is rarely filled. It becomes attractive only if curtailment structurally exceeds 70%, which is possible but not the base case.

15-Year Financial Projection

Option B (10 MW / 30 MWh) at 50% Curtailment (Current Park Baseline) — conservative scenario using constant curtailment rate and constant electricity prices.

Year	PV Prod. (MWh)	SOH (%)	Eff. BESS (MWh)	Curtailed (MWh)	Captured (MWh)	Discharged (MWh)	Revenue (€)	O&M (€)	Net Income (€)	Cumulative (€)
1	19,324	94.62	28.39	9,662	8,247	7,119	1,302,691	89,088	1,213,603	1,213,603
2	19,228	92.66	27.80	9,851	8,669	7,484	1,369,253	89,088	1,280,165	2,567,098
3	19,132	90.70	27.21	9,801	8,625	7,446	1,362,302	89,088	1,273,214	3,840,312
4	19,036	88.74	26.62	9,752	8,972	7,745	1,417,358	89,088	1,328,270	5,168,582
5	18,941	86.78	26.03	9,704	8,928	7,707	1,410,406	89,088	1,321,318	6,489,900
6	18,846	85.34	25.60	9,655	8,883	7,668	1,403,272	89,088	1,314,184	7,804,084
7	18,752	83.90	25.17	9,607	8,838	7,630	1,396,320	89,088	1,307,232	9,111,316
8	18,658	82.46	24.74	9,559	8,761	7,561	1,383,693	89,088	1,294,605	10,405,921
9	18,565	81.02	24.31	9,511	8,608	7,430	1,359,731	89,088	1,270,643	11,676,564
10	18,472	79.58	23.87	9,463	8,453	7,297	1,335,405	89,088	1,246,317	12,922,881
11	18,380	78.39	23.52	9,416	8,329	7,190	1,315,818	89,088	1,226,730	14,149,611
12	18,288	77.20	23.16	9,369	8,201	7,079	1,295,510	89,088	1,206,422	15,356,033
13	18,196	76.01	22.80	9,322	8,074	6,970	1,275,567	89,088	1,186,479	16,542,512
14	18,105	74.82	22.45	9,276	7,948	6,861	1,255,624	89,088	1,166,536	17,709,048
15	18,014	73.61	22.08	9,229	7,818	6,749	1,234,960	89,088	1,145,872	18,854,920

€18.10M

15-YEAR TOTAL
NET INCOME

4.48x

RETURN ON
INVESTMENT

3.3 yr

PAYBACK
PERIOD

€1.21M

AVG ANNUAL
NET INCOME

Conservative Model

This projection assumes *constant* 50% curtailment (the park's current baseline) and *constant* electricity prices throughout the 15-year period. In reality, both curtailment rates and evening peak prices are trending upward, meaning actual returns are likely to exceed this model. The payback milestone is reached during Year 4 (highlighted row), after which all net income is pure profit.

SOH Degradation Notes

Battery state-of-health (SOH) is modelled at 1 cycle/day, 0.25P C-rate using Linyang's LFP degradation curve. At Year 10, the battery retains 79.58% of original capacity (23.87 MWh effective from 30 MWh nominal). At Year 15, it retains 73.61% (22.08 MWh effective). These are OEM-warranted values under the LTSA agreement.

Future Revenue Opportunities

Important: Not Included in Base Financial Model

The following revenue streams are *not* included in the financial projections above. They represent additional upside that could materially increase returns once Cyprus opens ancillary services and grid-balancing markets to BESS operators.

Frequency Containment Reserve (FCR)

€200K – €600K/yr

Rate: €10–15/MW/hr

Hours: 2,000–4,000 hrs/yr

Calculation: 10 MW × €10–15/MW/hr × 2,000–4,000 hrs

Primary frequency response — fast injection/absorption to stabilise grid frequency. Cyprus TSO is developing this market as part of EU harmonisation.

Automatic Frequency Restoration (aFRR)

€50K – €300K/yr

Rate: €5–10/MW/hr

Hours: 1,000–3,000 hrs/yr

Calculation: 10 MW × €5–10/MW/hr × 1,000–3,000 hrs

Secondary frequency response — automated restoration of frequency deviations. Expected to launch on Cyprus grid by 2027–2028.

Grid Arbitrage (DAM Trading)

Up to €516K/yr

Spread: €81.86/MWh (peak – midday)

Net after RTE: €70.66/MWh

Calculation: €70.66 × 20 MWh × 365 days (1 cycle/day)

Buy cheap midday, sell at evening peak. Only available on non-curtailment days — curtailment recovery is always more profitable (€157.96 vs €70.66 per MWh stored).

Combined Upside Potential

€250K – €1.4M/yr

On top of curtailment recovery revenue

Priority order:

1. Curtailment recovery (always first)
2. FCR (stackable, doesn't require cycling)
3. aFRR (partially stackable)
4. Grid arbitrage (non-curtailment days only)

Market Timing

Cyprus is an EU member state and must harmonise its electricity market rules with the EU Clean Energy Package. This includes opening ancillary services (FCR, aFRR, mFRR) to storage assets. The TSO (TSOC) has indicated a phased rollout beginning 2027. BESS assets installed now will be positioned to capture these markets from day one. As of February 2026, BESS systems cannot buy from the DAM grid — all revenue in the base model is curtailment recovery only.

Methodology & Assumptions

Data Sources

Electricity Prices	TSOC Day-Ahead Market (DAM) — 134 consecutive days (Oct 2025 – Feb 2026)
PV Production	PVSyst simulation verified against 18 months of actual meter data (May 2024 — Oct 2025) — 19,324 MWh/yr, 1,932 kWh/kWp
Curtailment Data	Actual DSO dispatch-down records for Spanercom Anarita EAST + WEST arrays (May 2024 — Oct 2025)
BESS Pricing	Linyang Energy commercial offer (CIF Limassol), March 2026
O&M Rates	Linyang LTSA Tier C quotation (OEM-confirmed)
SOH Degradation	Linyang LFP degradation model (1 cycle/day, 0.25P C-rate)

Key Assumptions

Parameter	Value	Notes
Discharge Price	€182.99/MWh	Weighted average peak price (17:00–21:00), 134-day dataset
Charge Cost	€0/MWh	Curtailed energy — would be lost without BESS
RTE (AC-AC)	86.32%	Full system: battery + PCS + transformer + auxiliaries
Cycling Regime	1 cycle/day	Charge during curtailment, discharge at evening peak
BESS Availability	97%	LTSA-guaranteed minimum availability
PV Degradation	0.5%/yr	Linear degradation from Year 1
Price Escalation	0%/yr	Conservative — no inflation applied to electricity prices
O&M Escalation	0%/yr	Conservative — fixed LTSA rates assumed
Discount Rate	Not applied	Simple payback model (no NPV/IRR in base case)
Curtailment Trend	Constant	Held at scenario rate — actual trend is upward

Availability Factor

97% system availability is applied to both the throughput cap and the curtailment-limited capture formula, reducing annual energy by ~11 days of downtime.

Cycle Assumption

The model assumes one full charge/discharge cycle per day (annual average). Actual cycles vary seasonally: winter months (Dec–Jan) may average 0.7–0.8 cycles due to shorter solar days, while spring/summer may see daily curtailment exceeding BESS capacity. The annual average of 1 cycle/day is considered conservative.

Price Dataset Limitation

Peak discharge price of €182.99/MWh is based on a 134-day dataset (October 2025 – February 2026). This covers autumn and winter only — summer evening peak prices are not yet available from the TSOC DAM. The model assumes this price level remains constant year-round.

Data Provenance

PV production based on PVSyst simulation verified against 18 months of actual meter data (May 2024 — Oct 2025). Specific yield: 1,932 kWh/kWp.

Curtailment rates from actual DSO dispatch-down records for Spanercom Anarita EAST + WEST arrays. 2025 average (Jan–Oct): 48%, with spring peaks of 66–67%.

Capture Efficiency Model

Capture efficiency reflects the percentage of curtailed energy the BESS can actually store, accounting for curtailment timing variability:

Ratio (Daily Curtailed / BESS Capacity)	Capture Efficiency	Interpretation
> 1.5	97%	BESS always fills completely — excess curtailment
> 1.2	95%	Nearly always fills — strong oversupply
> 1.0	92%	Usually fills — moderate oversupply
> 0.8	88%	Often fills — some days under-curtailed
> 0.6	82%	Partially fills — variable curtailment
≤ 0.6	72%	Underutilised — curtailment often insufficient

Disclaimers

Important Notices

- This analysis is for informational and planning purposes only and does not constitute financial advice or a guarantee of returns.
- Electricity prices are based on historical data (Oct 2025 – Feb 2026) and may vary significantly. Past performance does not guarantee future results.
- Curtailment rates are estimates based on observed trends. Actual curtailment depends on TSO dispatch decisions, grid infrastructure development, and regulatory changes.
- Future revenue streams (FCR, aFRR, arbitrage) are estimates based on European market benchmarks and are subject to regulatory implementation timelines in Cyprus.
- The model does not account for financing costs, tax implications, grid connection fees, or permitting costs which may affect actual returns.
- All BESS pricing is CIF Limassol as quoted by Linyang Energy. Installation, civil works, grid connection, and commissioning costs are additional and covered in the separate commercial offer (Ref: LH-ANARITA-2026-03).

Prepared by: Lighthief Cyprus Ltd — EPC & Technical Advisory

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Reference: LH-FA-ANARITA-2026-03

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